Solutions

2. This question addresses induction machines

An induction machine is rated as follows: 415 V line; 50 Hz; 3-phase; 8-pole; star-connected. The equivalent circuit parameters, referred to the stator, are:

Stator resistance	1.0 Ω
Referred rotor resistance	0.5 Ω
Stator leakage reactance	2.0 Ω
Referred rotor leakage reactance	1.0 Ω
Magnetising reactance	350 Ω
Magnetizing resistance	350 Ω

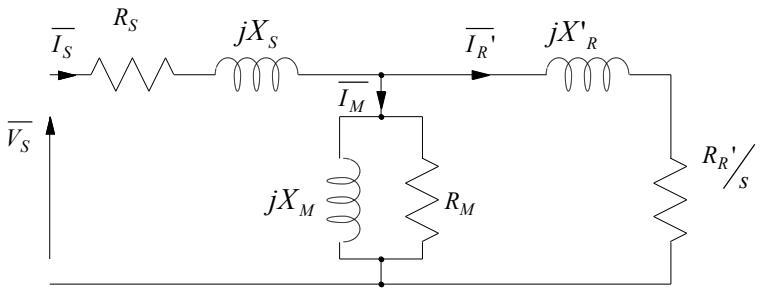
(a) Briefly and using some quantification justify ignoring the magnetising impedance for a slip of 2 % and indicate at what speed this approximation may not be valid. [2]

The Magnetising impedance has a magnitude of square root of 61250 = 247.5 – the referred rotor impedance has a magnitude of approx. 25 (dominated by $\frac{R_R'}{s} = 25$) so ten times – can ignore.

As the speed increases to closer to the synchronous speed and slip reduces then $\frac{R_R'}{s}$ gets larger and may reach a value where you cannot ignore the magnetizing impedance. For example, at a slip of 0.5 % $\frac{R_R'}{s}$ is 100.

(b) Ignoring the magnetizing impedance calculate the following for a slip of 2 %:

The equivalent circuit is:



(i) The speed in revolutions per minute [2]

$$\omega_S = {}^{\omega_E}/_P = 50 \times 2\pi/4 = 25\pi \frac{\text{rad}}{\text{s}} = 78.44 \frac{rads}{sec} = 750\ revs\ per\ minute$$

And rotor speed in revs per minute (RPM)

$$\omega_R = (1 - 0.02) * 750 = 735\ rpm$$

(ii) The stator current in Amps [2]

Rotor referred current is approximately equal to stator current assuming magnetizing impedance can be ignored

$Z_T = 1 + j2 + 25 + j2 = 26 + j4 = 26.31\angle 8.75°$

Stator current is calculated by: $I_S = \frac{V_L/\sqrt{3}}{Z_T} = \frac{400}{26.3}\angle -8.75° = 15.2 \angle -8.75°\text{A}$

(iii) The electromagnetic torque in NM [2]

$T_{EM} = \frac{3\, I_R'^2\, R_R'\left(\frac{1-s}{s}\right)}{\omega_R} = 110.39\ \text{Nm}$

(iv) The power factor (stating whether it is lagging or leading) [2]

Power Factor is given by angle of total input impedance:

$pf = \cos(-8.75) = 0.988\ lagging$

3. This question addresses transformers

A 50 Hz step-up transformer has the following parameters for its equivalent circuit

Primary winding resistance	R_1 = 1.0 Ω
Primary leakage reactance	X_1 = 5.0 Ω
Secondary winding resistance	R_2 = 0.2 Ω
Secondary leakage reactance	X_2 = 1.0 Ω
Magnetising resistance	R_M = 5,000 Ω
Magnetising reactance	X_M = 20,000 Ω

(a) The transformer primary is connected to a voltage of V_1 = 10 kV and the number of turns on one side of the transformer is 25 % the number of turns on the other side.

(i) Calculate the nominal secondary voltage [2]

$$V_2^{nom} = 4V_1^{nom} = 40{,}000\ kV$$

(ii) Calculate the voltage on the secondary side when it is open circuit [6]

$$V_2^{oc} = 4 * V_1 * \frac{(R_M \,/\!/\, jX_M)}{R_1 + jX_1 + (R_M \,/\!/\, jX_M)} = 4 * V_1 * \frac{(R_M \,/\!/\, jX_M)}{Z_T}$$

$$(R_M \,/\!/\, jX_M) = \frac{R_M \times jX_M}{R_M + jX_M} = \frac{5000 \times j20000}{5000 + j20000} = j10^8 \frac{5000 - j20000}{(5000)^2 + (20000)^2}$$

$$= 4705.88 + j1176.47 = 4850.713\angle 0.004276°$$

$$Z_T = 1 + j5 + 4705.88 + j1176.47 = 4706.88 + j1181.47 = 4852.9\angle 0.004292°$$

$$V_2^{oc} = 40000 \frac{4705.88 + j1176.47}{1 + j5 + (R_M \,/\!/\, jX_M)} = 40000 \frac{4850.713\angle 0.004276°}{4852.9\angle 0.004292°}$$

$$V_2^{oc} = 39981.99 - j0.01157 = 39981.99\angle -0.0000165732°$$

(iii) Calculate the magnetizing losses and the transformer efficiency when the transformer is open circuit on the secondary side. [4]

$$Magnetizing\ Losses = \frac{(\frac{39981.99}{4})^2}{5000} = 19.982\ kW$$

Efficiency is zero since output is open circuit.

(iv) If the transformer was operated at 60 Hz give an estimate of any increase or decrease in magnetizing losses (there is no need to recalculate the losses) and indicate what physical phenomena is responsible for any change. [4]

If the frequency goes up to 60 Hz then the hysteresis losses will increase by 20 % ((60-50)/50 or 4kw) as hysteresis losses are directly proportional to the frequency i.e. every time around the B-H curve is a loss – at 60 Hz we go faster around.

(b) Why were transformers fundamental to the outcome of the "current wars" i.e., AC versus DC in the late 19th century. [4]

When Thomas Edison tried to expand his Pearl Street DC power system to more and more consumers, he found that the voltage drops on the DC lines due to the resistance of the conductors to be detrimental to maintaining a certain voltage level, the voltage drop is proportional to the current times the resistance. The losses also increased and did so in a square law with respect to current and the current had to increase to supply more customers. This was truly a no-win situation. There was no way to address this other than to make the conductors lower resistance which would require larger cross-sectional area and hence very expensive. An alternative would be to have a higher voltage at the sending end but that would make the voltage too high near the generator etc. With an AC supply and a transformer, the voltage can be raised and the current reduced in direct inverse proportion to one another. So, increasing the voltage reduced the current which in turn reduced the losses (half current is a quarter the losses) and reduced the voltage drop.

5. This question addresses three-phase power systems

(a) What are the advantages of three phase power systems over single phase power systems. [4]

Three phases can easily produce a smooth rotating magnetic field which is useful for electrical machines. Single phase can only produce a pulsating field.

Three phase power systems can save a wire over single or two phase and hence costs (assuming they are balanced).

For three phase the instantaneous power is constant in a single phase it is pulsating.

On a distributed winding basis the voltage "lost" is minimal

(b) Consider a three-phase 60Hz balanced supply with a line voltage of 400 V. Three equal impedances are formed into a star connection and connected to the supply. The line current drawn is $200\sqrt{2/3} \angle +45^\circ \ A$. (The angle of the current is with respect to the phase voltage).

(i) Calculate the real and reactive powers drawn by the impedances. [3]

$$P = \sqrt{3} * 400 * 200 * \sqrt{2/3} * Cos\ (45°) = 80kW$$

$$Q = \sqrt{3} * 400 * 200 * \sqrt{2/3} * Sin\ (45°) = 80kVar$$

(ii) Calculate the impedances. [3]

Balanced so equal and regardless of whether it is 3 or 4 wire the following applies.

$$Z_A = Z_B = Z_C = \frac{V_A}{I_A} = \frac{400}{\sqrt{3}\ 200\sqrt{2/3} \angle +45°} = \sqrt{2} \angle -45°\Omega = 1 - \mathrm{j}$$

Aside: Not asked for here but useful later

$$\boldsymbol{I_A = 200\sqrt{2/3} \angle +45°\Omega}$$

$$\boldsymbol{I_B = 200\sqrt{2/3} \angle -75°\Omega}$$

$$\boldsymbol{I_C = 200\sqrt{2/3} \angle -195°\Omega}$$

The "The angle of the current is with respect to the phase voltage" is important here as the student may assume that the line current phase angle is with respect to a line voltage which is different.

If the angle was with respect to the line voltages, then the angle with respect to the phase voltage would be $\angle + 75°$ giving

$$\boldsymbol{Z_A = Z_B = Z_C = \frac{V_A}{I_A} = \frac{400}{\sqrt{3}\,200\sqrt{2/3}\angle + 75°}}$$

$$\boldsymbol{= \sqrt{2}\angle - 75°\Omega = 0.366 - j1.366}$$

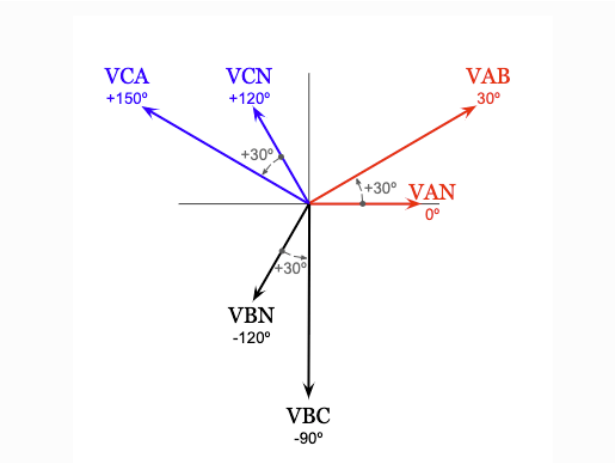


Figure: Not required but useful.

(iii) Calculate the value of the circuit element (inductor or capacitor) that can be placed in series with the A phase impedance so as its purely resistive, i.e., zero reactance. [4]

$$Z_A = 1 - j + Z_{add} = 1$$

$$Z_{add} = +j$$

$$\omega L = 1$$

$$L = \frac{1}{\omega} = \frac{1}{2 * \pi * 60} = \frac{1}{120\pi} = 2.65mH$$

(iv) With the A phase impedance purely resistive (as in (iii) above) and the B & C phase impedances unchanged the load is no longer balanced. Assuming an ABC sequence and a 4-wire configuration calculate all three phase and line currents and current in the neutral. [4]

Phase and line currents are the same as it is star connected and as it is 4-wire we can simply use the phase voltages and impedance values to calculate the currents. Also, B and C phase currents and line currents are not changed by this change in the A impedance.

$$I_A = \frac{400/\sqrt{3}}{1}\angle -0° = \frac{400}{\sqrt{3}} = 230.9 \text{ Amps}$$

$$I_B = \frac{400/\sqrt{3}\angle -120°}{\sqrt{2}\angle -45°} = 200\sqrt{2/3}\angle -75° = 42.265 - j157.735\ Amps$$

$$I_C = \frac{400/\sqrt{3}\angle -240°}{\sqrt{2}\angle -45°} = 200\sqrt{2/3}\angle -195° = -157.735 + j42.265\ Amps$$

$$I_N = I_A + I_B + I_C = 115.47 - j115.47 = 200\sqrt{2/3}\ \angle -45° Amps$$

(v) With the A phase impedance purely resistive (as in (iii) above) and the B & C phase impedances unchanged the load is no longer balanced. Assuming an ACB sequence and a 3-wire configuration calculate all three phase currents and the voltage at the star point.

No longer balanced and no longer four wires so not as straightforward as above - need to do some detailed calculations. Note phase sequence has changed so C & B switch in the Figure above – A stays where it is. The star point will no longer be at neutral important to illustrate this in a 3-wire case. Phasor diagram on graph paper would be ideal but they do not have graph paper so does not have to be too accurate.

$$V_{AC} = I_A * 1 - I_C * (1 - j) = 400\angle +30°$$

$$V_{AB} = I_A * 1 - I_B * (1 - j) = 400\angle -30°$$

$$I_A + I_B + I_C = 0$$

Three equations three unknowns

$$I_B = \frac{I_A - 400\angle -30°}{1 - j}$$

$$I_C = \frac{I_A - 400\angle +30°}{1 - j}$$

$$I_A + \frac{I_A - 400\angle -30°}{1-j} + \frac{I_A - 400\angle +30°}{1-j} = 0$$

$$I_A + \frac{I_A}{1-j} + \frac{I_A}{1-j} = \frac{400\angle +30° + 400\angle -30°}{1-j}$$

$I_A = \frac{400\angle+30°+400\angle-30°}{3-j} = \frac{692.82}{3-j}$ = 219.1∠18.43° = 207.85 + j69.28

I_B = 214.1∠162.23°= -203.92 + j65.36

I_C = 134.7∠ − 91.67°= -3.92 – j134.64

$I_N = 0$

Star voltage

$V_{SN} = V_{AN} - V_{AS} = \frac{400}{\sqrt{3}} - I_A = 23.1 - j69.28\, Volts$ [8]

(vi) For (v) above Sketch the phasor diagram showing all 3 phase currents and the voltage at the star point.

